

Element K: Reflection on the Design Project

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students' ability to rest.

First, we defined the problem statement as a team. We defined the problem well because we efficiently communicated with one another throughout the process. Then, we defined stakeholders' interests in their room: Student E, Student E's brother, and Ms. Z. We made this process efficient because Student E was a member of our team. This allowed us to visualize the problem. We started developing a viable solution by brainstorming the situation with what we had available to solve this issue. We brainstormed five sketches on paper and agreed on a good design. We developed a decision matrix to systematically compare solutions for the insect problem, soon after we had drawn sketches of our solutions. We identified critical design requirements, including effectiveness at capturing insects, safety, efficient use, assembly time, and insect attracting. The system also needed to be small and work effectively, so it noticeably reduces the insect population. Quick setup is also a requirement. Most importantly, the system must not harm pets, people, or the environment. Prioritizing the design requirements, set with feedback from our stakeholders, posed a challenge. We changed our prioritized design requirements numerous times due to some issues we faced throughout the project. Some of these issues include, but are not limited to, design requirements not being testable, design requirements being ineffective, and design requirements being unnecessary. We revised the design requirements to meet the needs of our stakeholders.

The lure for the insects in prototype one was a flashlight placed at the end of a paper towel roll, which had insect sticky paper inside the roll. The safety of the design was

reconsidered, and no people or pets were harmed in the making.^c Student E's test of the first physical prototype did not catch any insects while indoors at first. The team was unsure and found the test inconclusive. During the second test, outdoors, Student E had collected several insects. The team shifted the second test just to make sure that our prototype met our stakeholders' needs and caught insects.

Fortunately, our external expert identified an issue that our prototype has that could be hindering its effectiveness. The way the light is used in our prototype is similar to an attractant, while Dr. H says it blinds them instead of luring them to a trap. Materials were easily sourced for the prototype and were all quickly gathered for assembly. Unfortunately, Dr. H's feedback, while valuable, outlined a solution that was unrealistic and difficult to achieve. Our second physical prototype, when printed for testing, was printed at a fraction of the size, reducing its effectiveness too much to be used or even tested.

When you are just starting to think about ideas, make quick models using a simple 3D printer if you have one. Easy models can help you see if the size and shape are right and how the basic catching part might work. For example, you could make a simple box with sticky paper inside to see if it is a good size to catch insects. Once you have a good idea, build the main parts that catch the insects. If you have a special trap or opening, make a model of just that part. Test it to see if it actually catches insects well and if it breaks easily. This will help you make those parts better. Show your simple models to the people with the insect problem and ask them what they think. Observe how they interact with it and get their ideas on how to make it more effective. Getting their hands-on feedback is helpful.